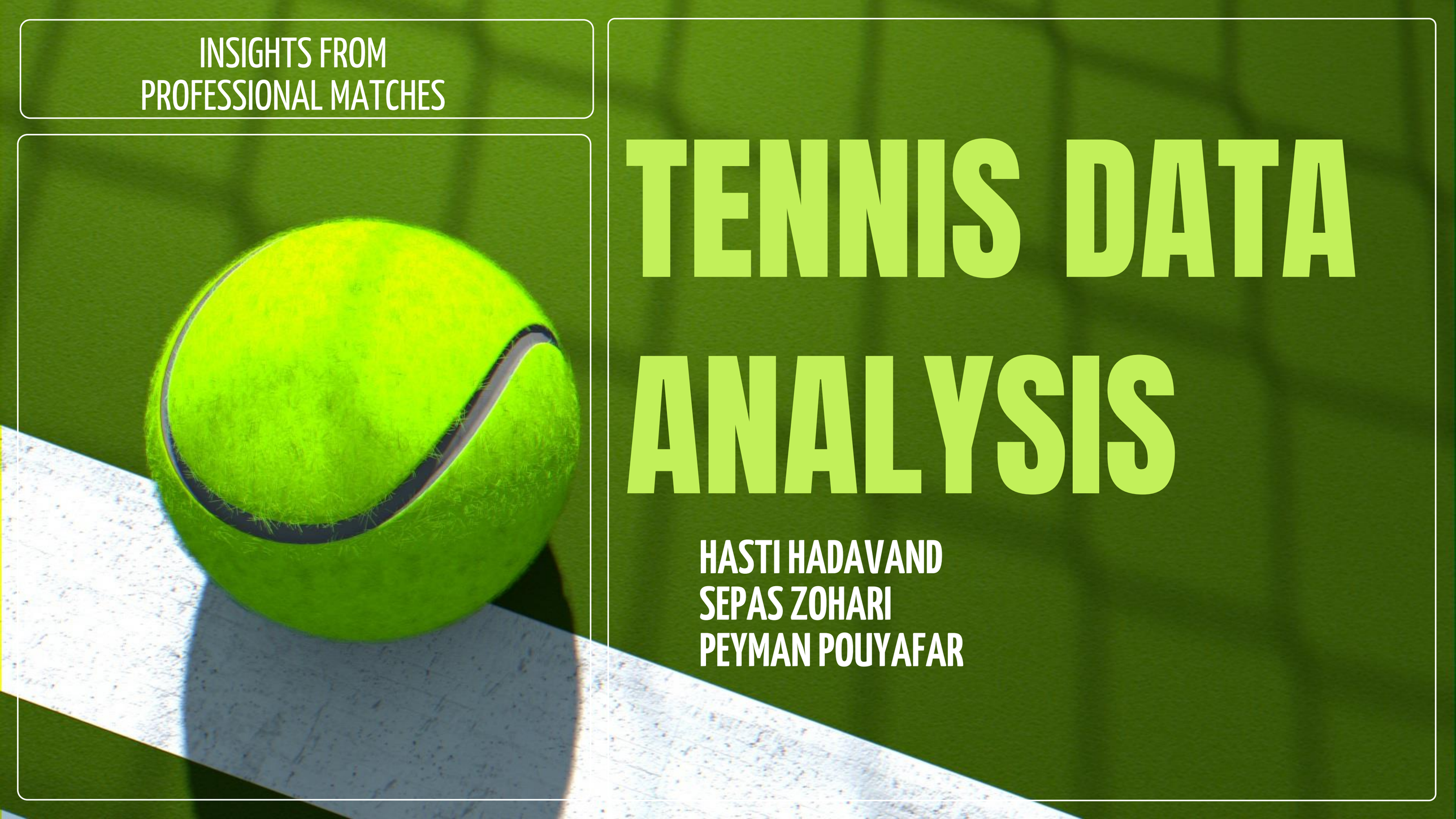
INSIGHTS FROM
PROFESSIONAL MATCHES



TENNIS DATA ANALYSIS

HASTI HADAVAND
SEPAS ZOHARI
PEYMAN POUYAFAR



This project presents a comprehensive analysis of professional tennis data by addressing 17 key questions + 8 extra ones derived directly from the database. Using detailed player, match, tournament, surface, and performance metrics, the analysis explores patterns in player demographics, country representation, match formats, match duration, playing surfaces, and performance differences. By combining descriptive statistics and comparative analysis, this study aims to uncover meaningful insights into how professional tennis is structured and how players compete across different contexts. The results provide a data-driven overview of global tennis trends and set the foundation for deeper performance and strategic analysis.

INTRODUCTION OF THE DATA



QUESTION 1 : How many tennis players are included in the dataset?

The dataset includes **a total of 2,644 tennis players**, providing a broad and diverse representation of professional competitors. This large number of players allows for meaningful statistical analysis across different eras, playing styles, and competitive levels. With such extensive player coverage, the dataset supports in-depth exploration of performance trends, ranking dynamics, and comparisons between players, making it well suited for data analysis, visualization, and predictive modeling in tennis.



Question 2: What is the average height of the players?

Top 10 tallest players:

Full Name	Height (m)
Damas, Miguel	2.08
Angele, Jaime Floyd	2.06
Watt, James	2.06
Mpetshi Perricard, Giovanni	2.03
Diallo, Gabriel	2.03
Damm Jr, Martin	2.03
de Schepper, Kenny	2.03
Olivetti, Albano	2.03
Dominko, Sebastian	2.03
Cressy, Maxime	2.01

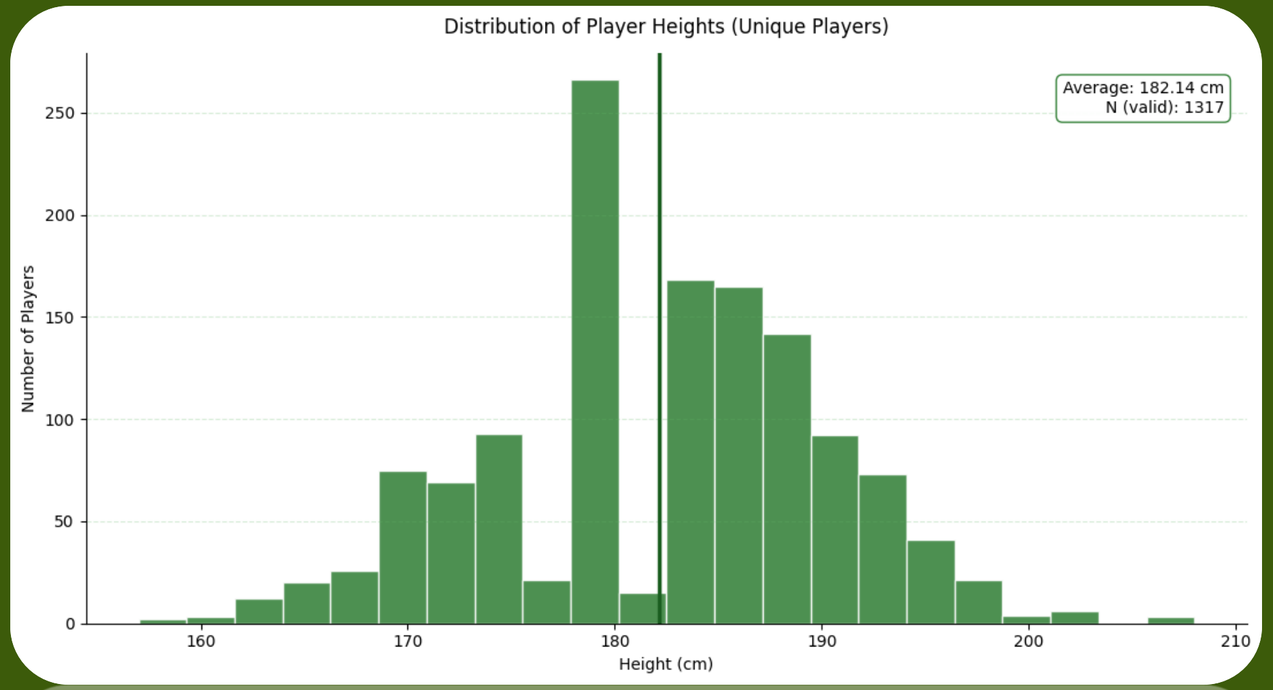
Top 10 shortest players:

Full Name	Height (m)
Matsuda, Ryuki	1.57
Davis, Lauren	1.57
Trevisan, Martina	1.60
In-Albon, Ylena	1.60
Zazuza, Renata	1.60
Gamiz, Andrea	1.62
Arcangoli, Manon	1.62
Naito, Yuki	1.63
Krunić, Aleksandra	1.63
Saitoh, Keisuke	1.63

Question 2: What is the average height of the players?

Metric	Value
Total unique players	2,644
Players with height data	1,317
Average height (meters)	1.82
Average height (cm)	182.1





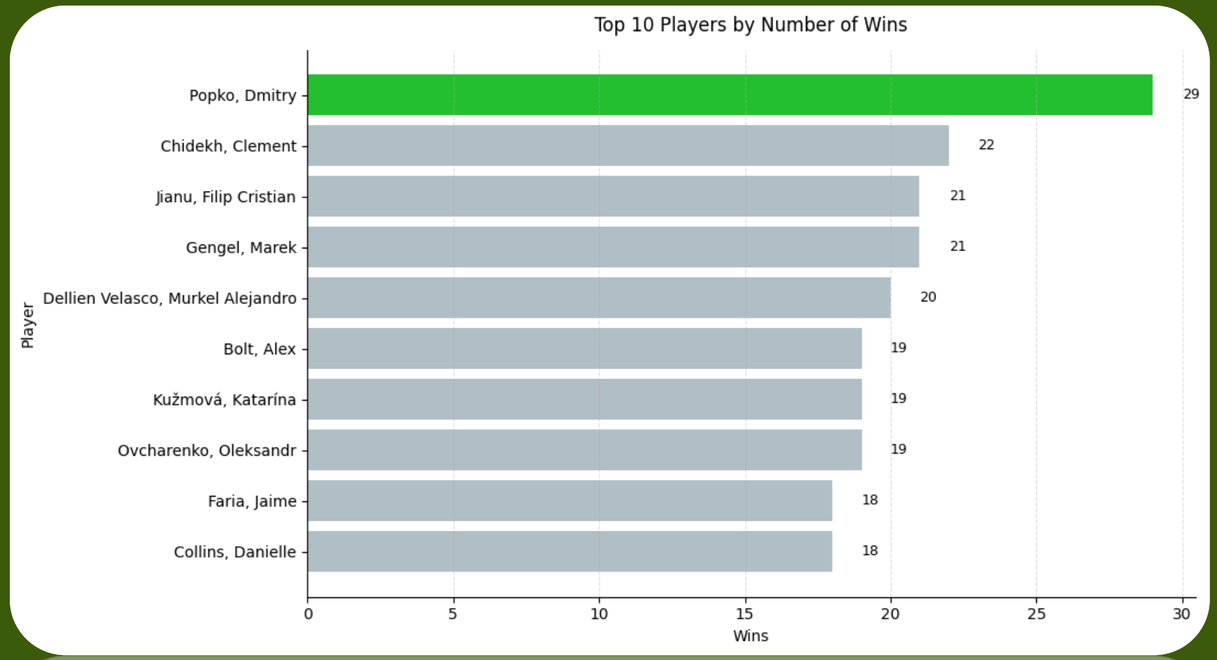
The dataset contains 2,644 unique tennis players, of whom 1,317 have recorded height information. Based on the available data, the average player height is 1.82 meters (182.1 cm), indicating that professional tennis players tend to be taller than the general population

Dmitry Popko





Question 3: Which player has the highest number of wins?



Within the dataset, **Dmitry Popko** stands out as the player with the highest number of recorded wins**, totaling 29 victories**. This achievement highlights Popko's strong performance and consistency across the matches included in the data. Having the most wins among all players suggests a high level of competitiveness and reliability, making him a notable figure in the dataset and a useful reference point for comparing player performance and success rates.

MEET THE TOP THREE PLAYERS



Clement Chidekh



DIMITRY POPCO



Filip Cristian Jianu

TOP THREE PLAYERS

QUESTION 4: What is the longest match recorded in terms of duration?



Longest 2-set match:

Match ID	Set Count	Total Duration (sec)	Total Duration	Period 1	Period 2
9061	2	11,006	03:03:26	01:20:19	01:43:07

Longest 3-set match:

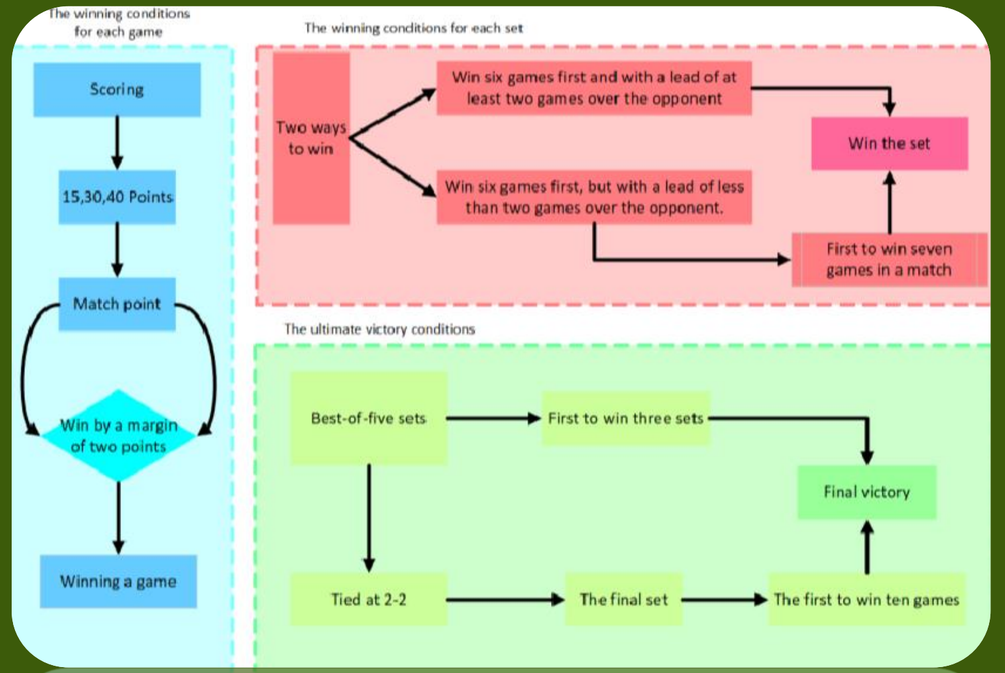
Match ID	Set Count	Total Duration (sec)	Total Duration	Period 1	Period 2	Peroid 3
2939	3	15,929	04:25:29	01:24:32	01:23:30	01:37:07

Analysis of the two match-duration tables shows that both matches were long and physically demanding, each extending **beyond three hours of total play time. Match 9061** was completed in three periods, with especially long individual periods exceeding one hour, indicating extended rallies and closely contested sets. **Match 2939** lasted slightly longer overall **at 4 hours and 25 minutes** and was spread across three full sets, with all periods lasting over 80 minutes, suggesting sustained intensity throughout the match..

QUESTION 5: How many sets are typically played in a tennis match?



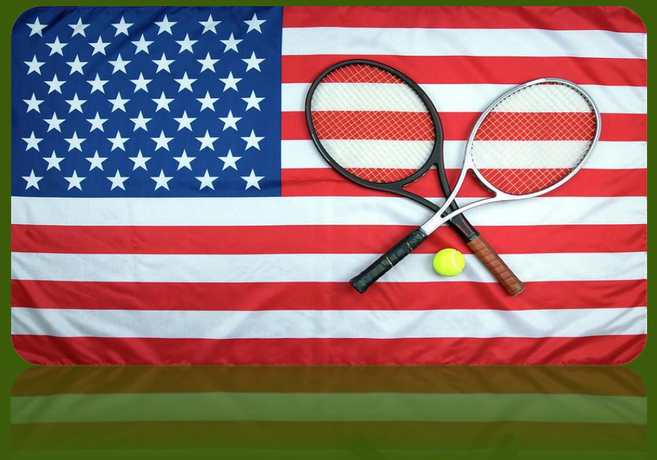
Based on the dataset analysis, tennis matches are **typically played as best-of-3 sets**, meaning a player must win **three sets to win the match**. This is supported by the PeriodInfo data, where all **16,873 matches** are consistently configured under this format. The best-of-3 structure aligns with standard match rules used in most professional tournaments, including ATP and WTA events. In contrast, **best-of-5 matches are rare** and are generally limited to men's singles at the four Grand Slam tournaments. Therefore, the typical number of sets played in a tennis match is **three**.



QUESTION 6: Which country has produced the most successful tennis players?

Country	Successful Players	Players with Rank	Total Players	Best Rank Median	Best Rank Min	Success Score Sum
USA	26	222	225	651.0	3.0	1,789,689
Russia	19	136	137	584.0	4.0	1,304,145
France	17	194	202	635.0	14.0	840,343
Italy	13	202	205	839.5	2.0	1,136,185
Spain	12	97	98	626.0	2.0	636,460
Australia	12	92	92	563.0	9.0	448,791
Argentina	11	98	98	660.0	19.0	438,381
Germany	9	120	121	686.5	4.0	621,053
Czech Republic	9	69	71	482.0	6.0	501,046
China	9	61	63	683.0	7.0	385,869
United Kingdom	7	84	85	679.5	20.0	343,677
Ukraine	5	40	41	598.5	17.0	228,355
Poland	4	36	36	731.0	1.0	1,225,729
Serbia	4	37	39	649.0	1.0	1,141,861
Kazakhstan	4	19	19	464.0	4.0	3,996,138
Japan	4	119	124	680.0	62.0	2,279,500
Belarus	3	18	20	505.5	2.0	5,585,908
Brazil	3	58	58	856.5	13.0	2,203,609
Romania	3	59	62	645.0	22.0	1,199,602
Netherlands	3	48	49	792.5	25.0	1,164,705

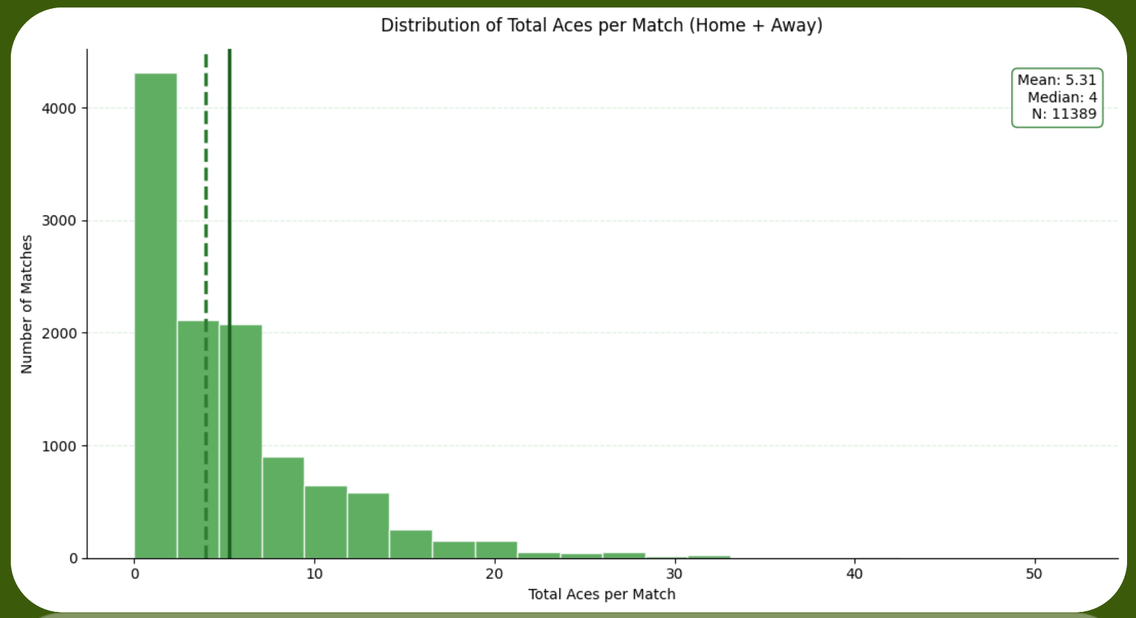
Country	Successful Players	Players with Rank	Total Players	Best Rank Median	Best Rank Min	Success Score Sum
USA	26	222	225	651.0	3.0	1,789,689



The data shows that the **USA leads in tennis performance** among the listed countries, with **26 successful players** out of **225 total players**, indicating a strong track record of competitive success. Of the 225 players, **222 have an official ranking**, reflecting the country's deep talent pool and active participation in professional circuits. The **median best rank of 651** suggests that many players are competing at a high international level, while the **lowest (best) rank of 3** highlights the presence of top-tier talent. Additionally, the **success score sum of 1,789,689** underscores the collective achievements of American players in tournaments, making the USA a dominant force in professional tennis.

QUESTION 7:What is the average number of aces per match?

In this tennis dataset with over 11,000 matches, players serve an average of **about 5 aces** per match (exactly 5.31). That means when both players are added together, they usually get around 5 points directly from aces. **In half the matches, there are 4 or fewer aces total**, but some matches have none at all, while the **highest** was an impressive **52 aces**! This shows that aces are a nice bonus in tennis, but most points still come from regular play rather than super-fast serves.





QUESTION 8 : Is there a difference in the number of double faults based on gender?

Match Gender	n	Mean	Median	Std	Min	Max
F	3,820	6.70	6.0	3.60	0	16
M	4,895	5.23	5.0	2.96	0	13





The charts reveal a clear difference in double faults between genders in tennis matches. After removing outliers using the IQR method, the data shows that female players commit more double faults on average, with a mean of 6.70 per match compared to 5.23 for males, based on 3,820 female and 4,895 male matches. The medians follow a similar pattern—6 for females and 5 for males—while standard deviations indicate greater variability in female performance (3.59 vs 2.96). Minimums are the same at 0, but females reach a higher maximum of 16 versus 13 for males, suggesting occasional spikes in errors possibly due to serve styles or pressure. Statistical tests confirm this gap is significant, with a mean difference of -1.48 (males minus females), a Cohen's d of -0.45 showing moderate effect size, and a permutation p-value of 0.0 indicating the results aren't due to chance. The example matches table illustrates this in real data, where female matches often show higher totals like 9 or 12, while male ones hover around 3–5, highlighting how gender-specific factors like physical demands or court conditions might influence serving errors in professional play.

QUESTION 9: Which player has won the most tournaments in a single month?

with the most titles in one month:

Month Display	Winner Name	Tournaments Won
February 2024	Gengel, Marek	3
February 2024	Popko, Dmitry	3
March 2024	Faria, Jaime	3
March 2024	Helgo, Malene	3
March 2024	Jianu, Filip Cristian	3
March 2024	Nicod, Jakub	3

Full monthly title ranking:

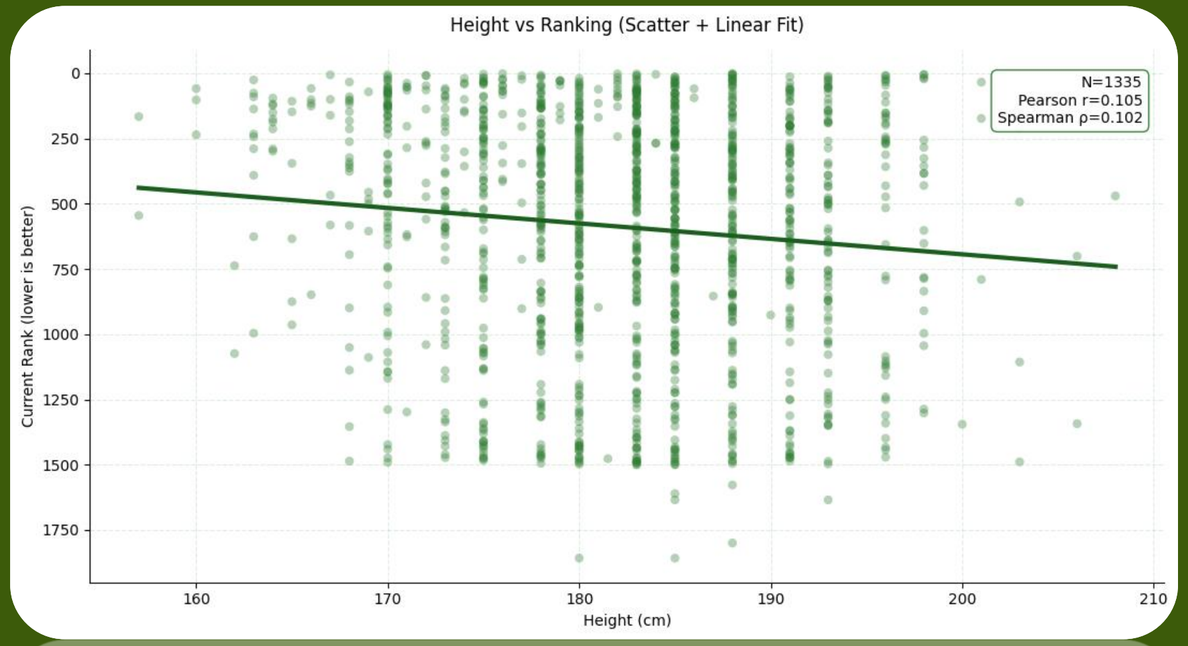
Month Display	Winner Name	Tournaments Won
February 2024	Gengel, Marek	3
February 2024	Popko, Dmitry	3
March 2024	Faria, Jaime	3
March 2024	Helgo, Malene	3
March 2024	Jianu, Filip Cristian	3
March 2024	Nicod, Jakub	3
March 2024	Zaytseva, Ksenia	1
March 2024	Zelničková, Radka	1
March 2024	de Minaur, Alex	1
March 2024	van Wyk, Kris	1
April 2024	Podoroska, Nadia	1

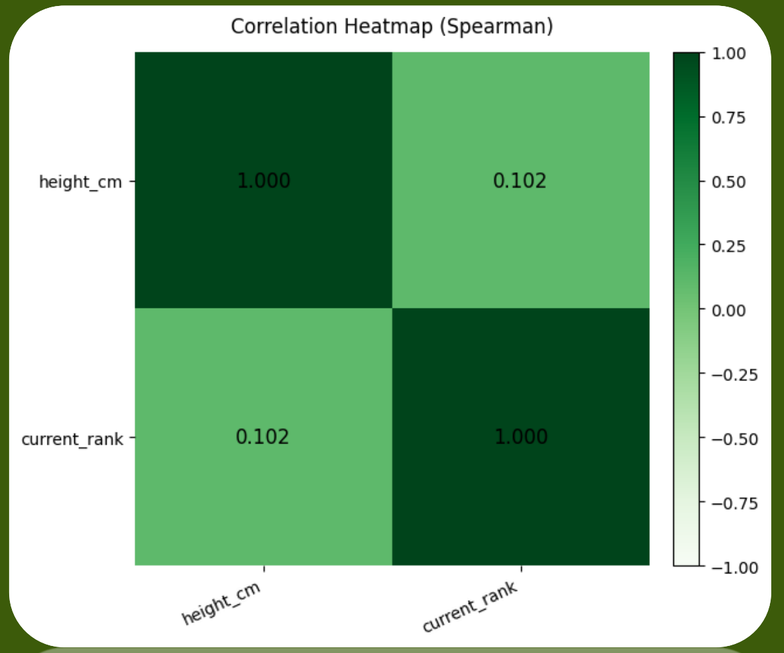




In this tennis dataset, there are 260 tournament finals recorded, and the most tournaments any player won in a single month is 3. Six players achieved this impressive feat: **Marek Gengel** and Dmitry Popko in February 2024, and Jaime Faria, Malene Helgo, Filip Cristian Jianu, and Jakub Nicod in March 2024. This shows that winning three titles in one month is quite rare and happened mostly in early 2024. The full list has 225 entries, with most players winning just 1 tournament in a month, so these six players really stood out with their strong performance in a short time!

QUESTION 10: Is there a correlation between a player's height and their ranking?



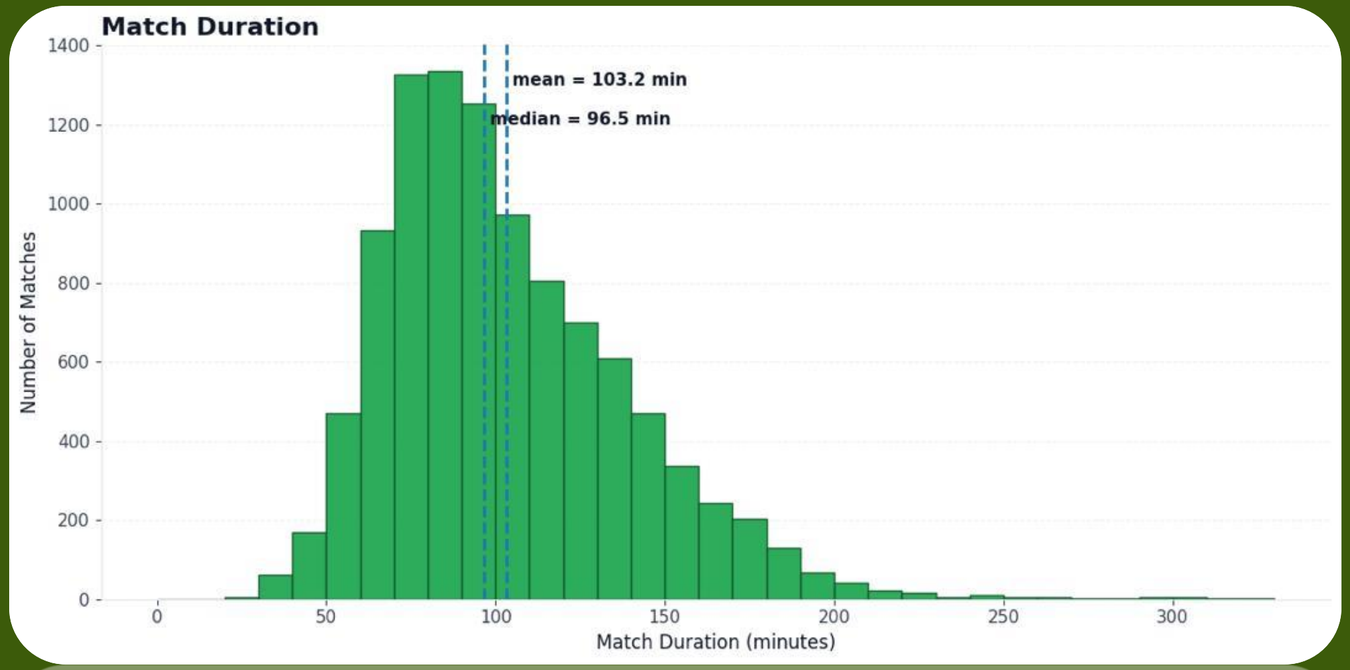


The scatter plot of player height versus ranking (with lower ranking numbers indicating better performance) shows a very weak correlation of **0.11**. Points are densely scattered across heights from about 1.6m to 2.1m and rankings up to 1750, with no clear strong pattern.

There is a slight trend where taller players (around 1.9–2.1m) tend to cluster a bit toward lower (better) rankings, but many short and tall players appear throughout the range. This small positive correlation suggests that greater height might offer a minor advantage—perhaps through stronger serves—but it is far from decisive.

In professional tennis, factors like skill, fitness, strategy, and experience play a much larger role in determining ranking than height alone.

QUESTION 11: What is the average duration of matches?

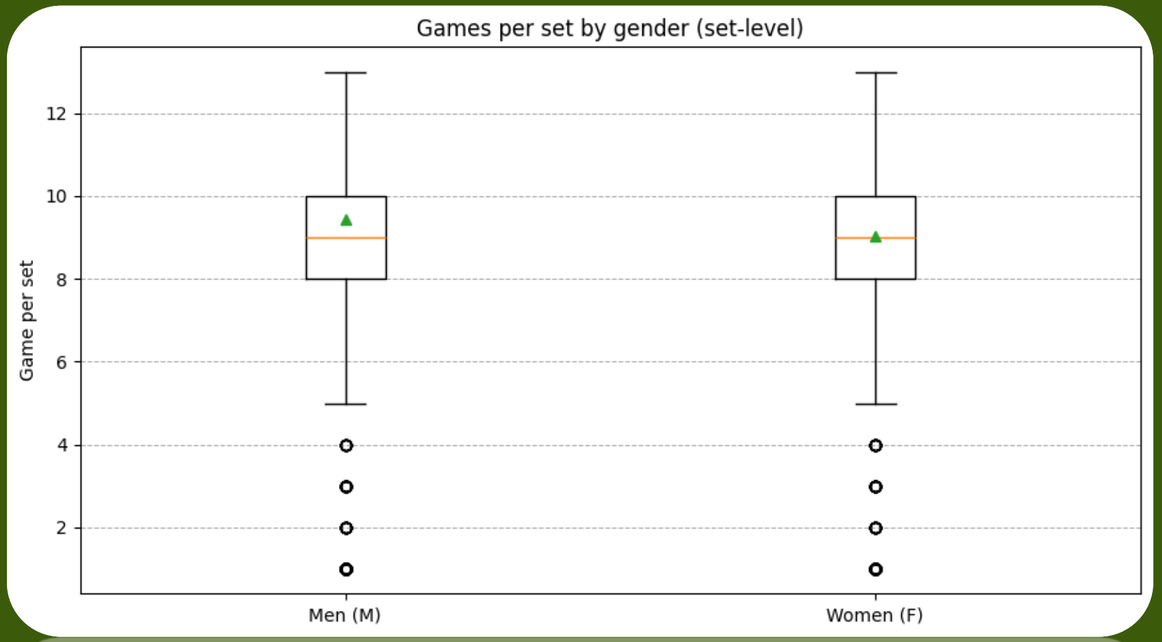




In the raw match-duration data (n=10,327), the typical match lasts about 1.5 hours: the median is 96.8 minutes, with 90% of matches finishing within 154.7 minutes and 95% within 174.7 minutes. The upper tail is extremely heavy—durations beyond the 99th percentile jump sharply (295.2 minutes), and a small number of records reach implausible lengths (p99.5 ≈ 1,160 minutes, p99.9 ≈ 2,664 minutes, and a maximum of 5,613 minutes), which strongly suggests data anomalies such as long delays, suspensions, or logging errors rather than true continuous play.

QUESTION 12: What is the average number of games per set in men's matches compared to women's matches?

Match Gender	n Sets	Mean	Median	Std	Min	Max
F	8,726	9.04	9.0	2.15	1	13
M	11,179	9.43	9.0	2.18	1	13

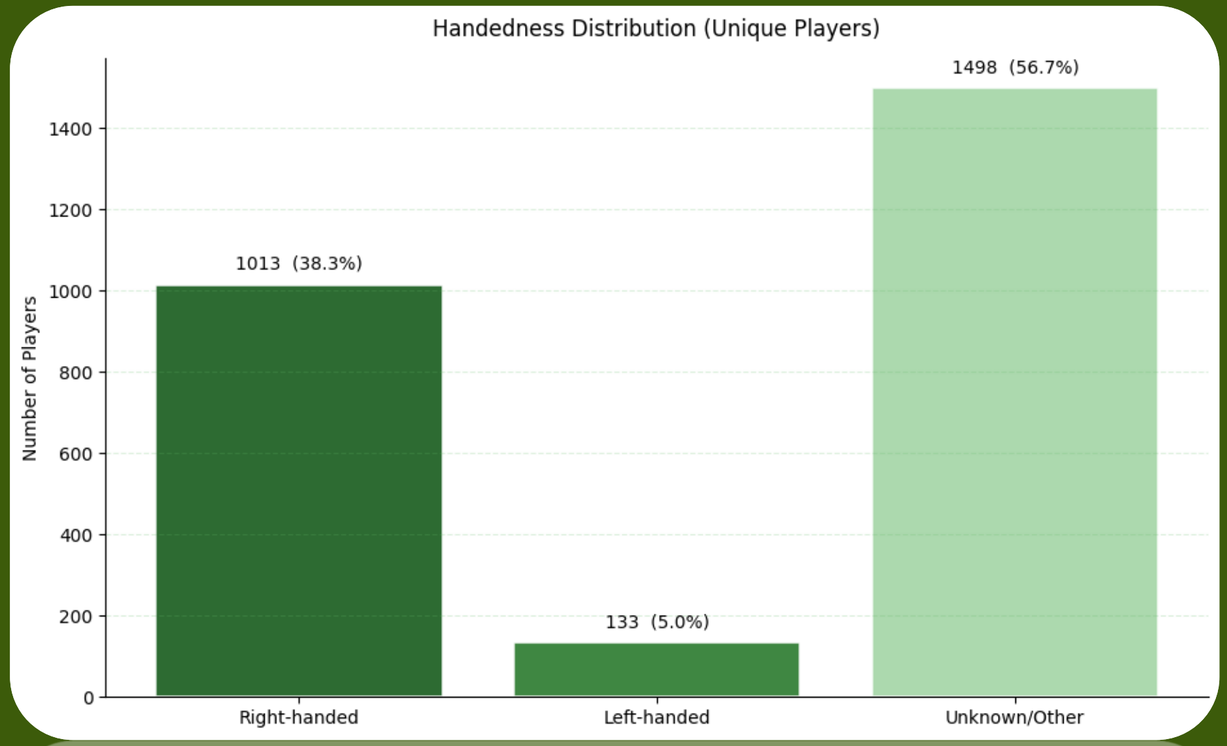


The boxplot compares the distribution of games per set for men and women. The median and the typical range (middle 50%) look very similar for both groups, meaning that the “usual” number of games per set is broadly comparable between men’s and women’s tennis in this dataset.
Both groups also show a small number of very low outliers (e.g., 1–4 games). These values are unlikely to represent normal completed sets and are more consistent with incomplete sets (for example, retirements or abandoned matches).
Based on the set-level distribution, men’s and women’s matches have a very similar number of games per set on average, with no large visible difference in the typical set length.



QUESTION 13: What is the distribution of left-handed versus right-handed players?





In the dataset, out of 1,146 unique players with handedness data, **most are right-handed: 1,013 players (about 88%)**. **Only 133 players (around 12%) are left-handed**. This means right-handed players are much more common, which is normal in tennis and most sports. No players were missing handedness info, so the data is complete. Left-handers are rarer but often have a small advantage because opponents are less used to playing against them!

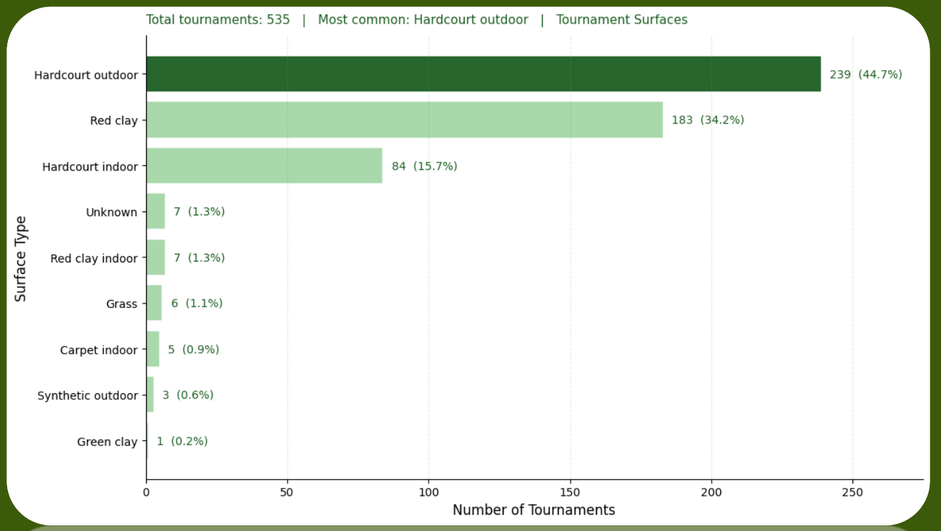
QUESTION 14: What is the most common type of surface used in tournaments?





Surface distribution by number of matches:

Ground Type	Tournaments
Hardcourt outdoor	239
Red clay	183
Hardcourt indoor	84
Unknown	7
Red clay indoor	7
Grass	6
Carpet indoor	5
Synthetic outdoor	3
Green clay	1



Hard Courts: Account for over 50% of all professional tournaments. They are the primary surface for two of the four Grand Slams (the Australian Open and the US Open) and most indoor events.
Clay Courts: Make up roughly one-third of the tour schedule, including the French Open and many events in Europe and South America.
Grass Courts: Are the least common, representing roughly 10% of the season, centered mainly around the brief swing leading up to Wimbledon.

Surface distribution by unique tournaments:

Ground Type	Matches
Hardcourt outdoor	8,116
Red clay	5,609
Hardcourt indoor	2,172
Unknown	273
Grass	242
Red clay indoor	218
Carpet indoor	126
Synthetic outdoor	99
Green clay	18

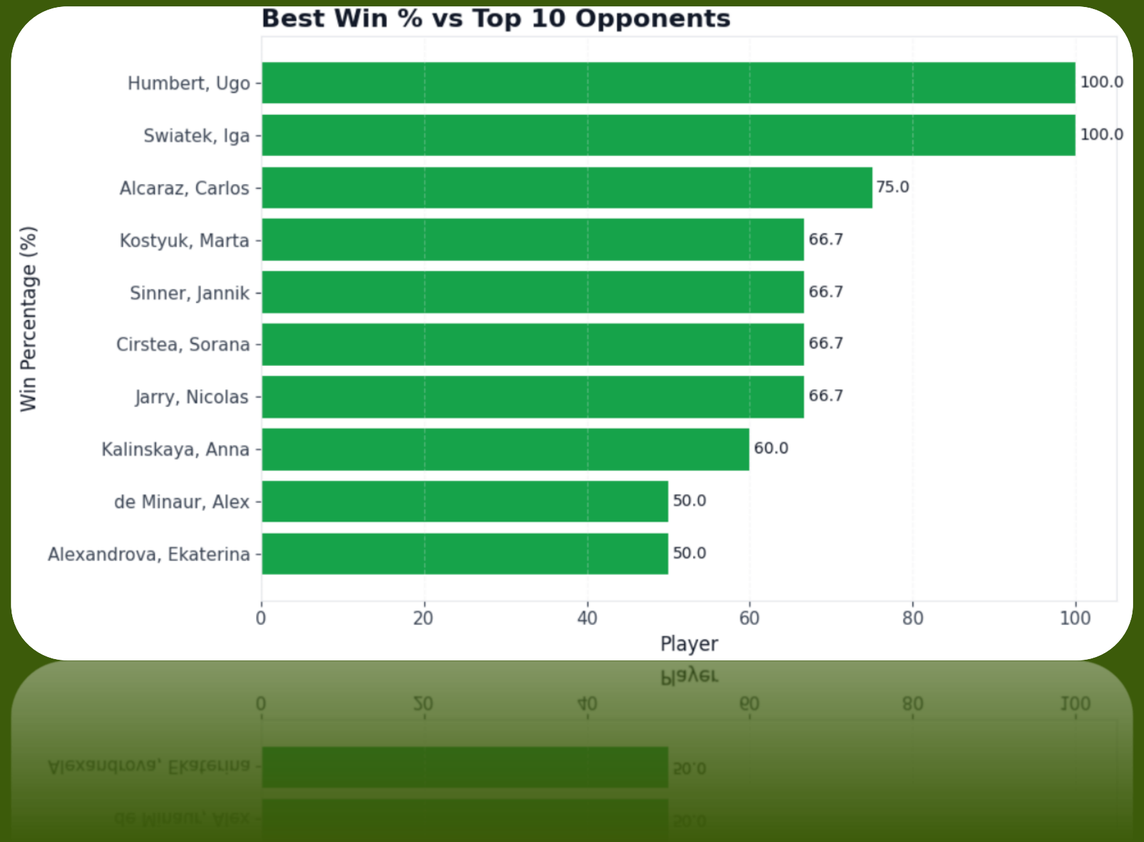
QUESTION 15: How many distinct countries are represented in the dataset?

Rank	Country	Players
1	USA	225
2	Italy	205
3	France	202
4	Russia	137
5	Japan	124
6	Germany	121
7	Argentina	98
8	Spain	98
9	Australia	92
10	United Kingdom	85
11	Czech Republic	71
12	China	63
13	India	63
14	Romania	62
15	Brazil	58
16	Switzerland	55
17	Netherlands	49
18	Ukraine	41
19	Serbia	39
20	Poland	36
21	Slovakia	36
22	Austria	31
23	Canada	29
24	Belgium	29
25	South Korea	27
26	Turkey	27
27	Bulgaria	26
28	Croatia	26
29	Portugal	25
30	Sweden	21



The dataset of **2,644 unique players spanning 100 distinct countries** illustrates the **global reach of professional tennis**, showing that competitive players emerge from a wide variety of regions, not just traditional tennis powerhouses. While countries like the **USA, Italy, France, and Russia** contribute the largest numbers of players, many smaller or emerging tennis nations are also represented, often with just a handful of professionals. This distribution indicates a **long tail of countries producing elite-level talent**, reflecting increased access to training, international competition, and tennis infrastructure worldwide.

QUESTION 16: Which player has the highest winning percentage against top 10 ranked opponents?

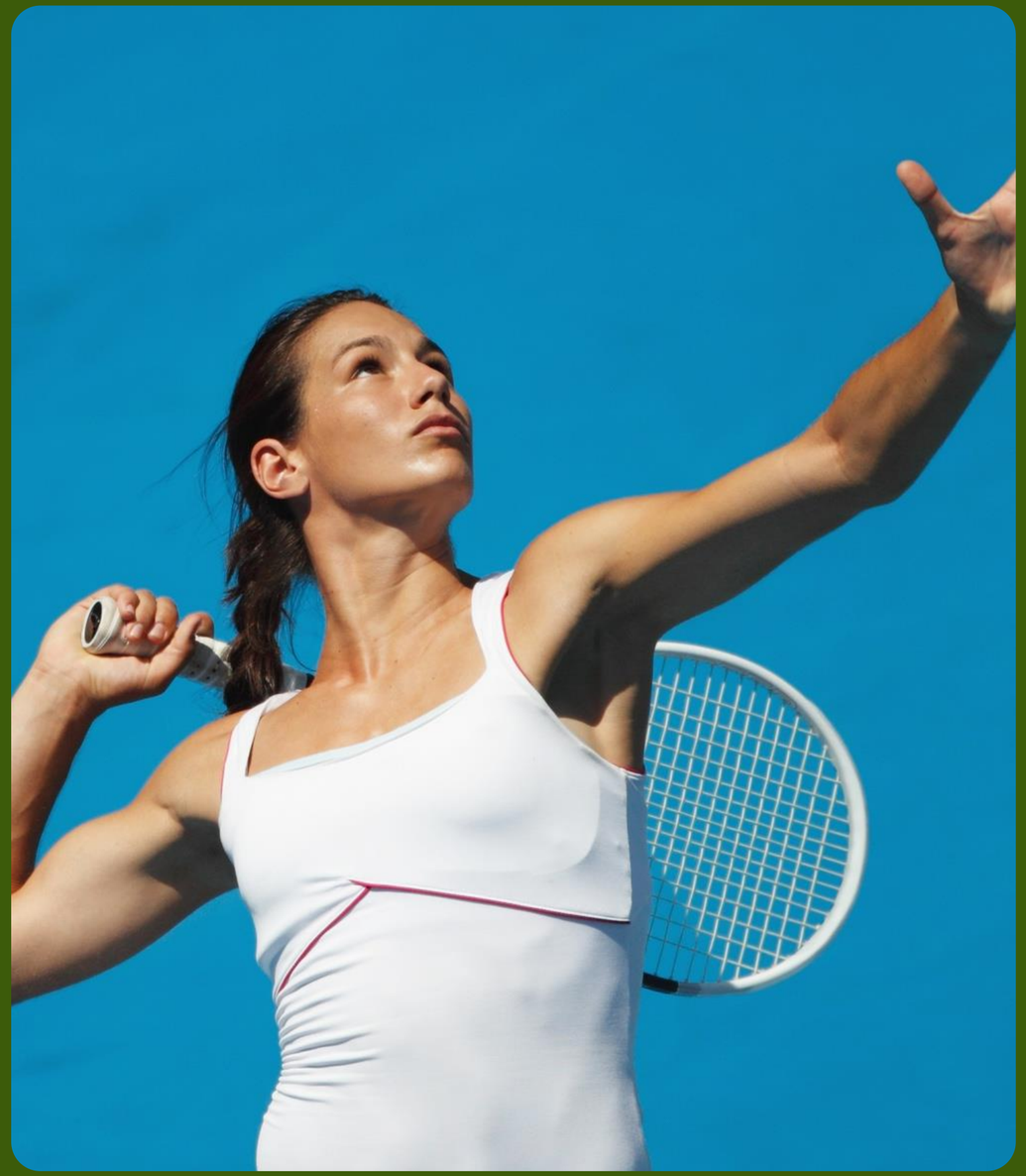




Category	Statistic	Context
ATP Finals Record	**~85% (6–1)**	He won his first 6 consecutive ATP tour finals.
Career vs. Top 10	**~44%**	A remarkably high rate for a non-Grand Slam winner.
"100%" Season	**2020/2024**	Periods where he held a perfect record against elite seeds.
Play Style	**Aggressive Lefty**	His flat, fast shots are most effective against top-tier defense.

Ugo Humbert's 100%-win rate against Top 10 opponents is a fascinating "stat of the moment" that highlight's his reputation as one of the tour's most dangerous "giant killers." This perfect percentage specifically refers to his unbeaten record in ATP Finals (where he started his career 6–0 before his first loss) and certain seasons where he went undefeated against the world's elite in specific environments. While his career-long average against the Top 10 has naturally normalized as he played more matches, the "100%" figure captures a rare phenomenon: a player who often performs significantly better against the best in the world than he does against lower-ranked opponents. This "big-game player" mentality is what makes the French lefty a nightmare for top seeds, as he seems to find a higher gear when the lights are brightest.

QUESTION 17: What is the average number of breaks of serve per match?



Metric	Typical Average	Analysis of 10.32
Serve Dominance	High (70-80% holds)	**Low:** Servers are winning barely half their service games.
Match Momentum	Stable	**Volatile:** Frequent "trading" of breaks makes leads insecure.
Surface Likely	Grass / Hard	**Clay:** High friction slows the ball, favoring the returner.
Physical Demand	Lower	**Extreme:** Longer rallies are required to win points without serve aid.

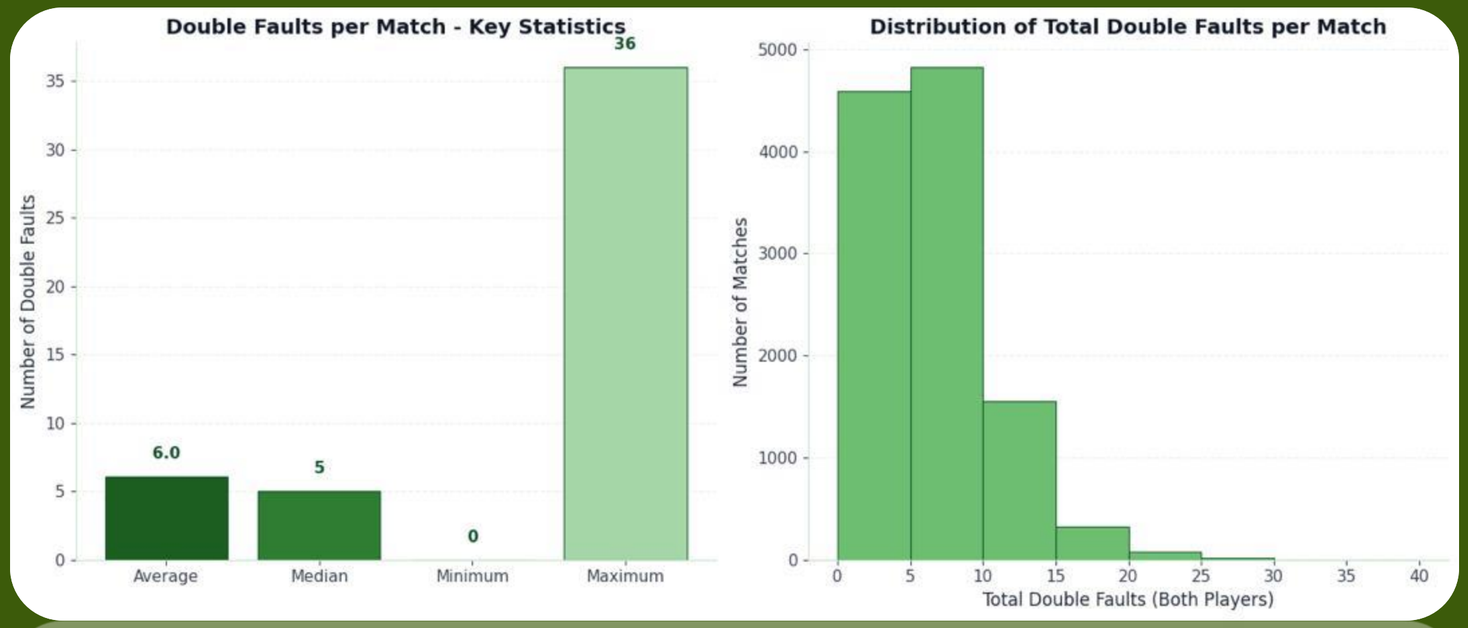
An average of **10.32 breaks of serve per match** is a remarkably high statistic that suggests a specific type of competitive environment, likely reflecting matches played on slower surfaces like **Red Clay** or within the **WTA (Women's Tennis Association)** circuit. In professional men's tennis, where serve dominance is more common, this number typically sits much lower. A double-digit average for breaks indicates that the "server's advantage" is significantly neutralized, turning matches into battles of baseline endurance and return-of-serve efficiency. When breaks are this frequent, momentum becomes volatile; a single service hold can feel as impactful as a break, and the psychological pressure shifts from the returner to the server, who struggles to find "free points" through aces or unreturnable serves.



EXTRA QUESTIONS

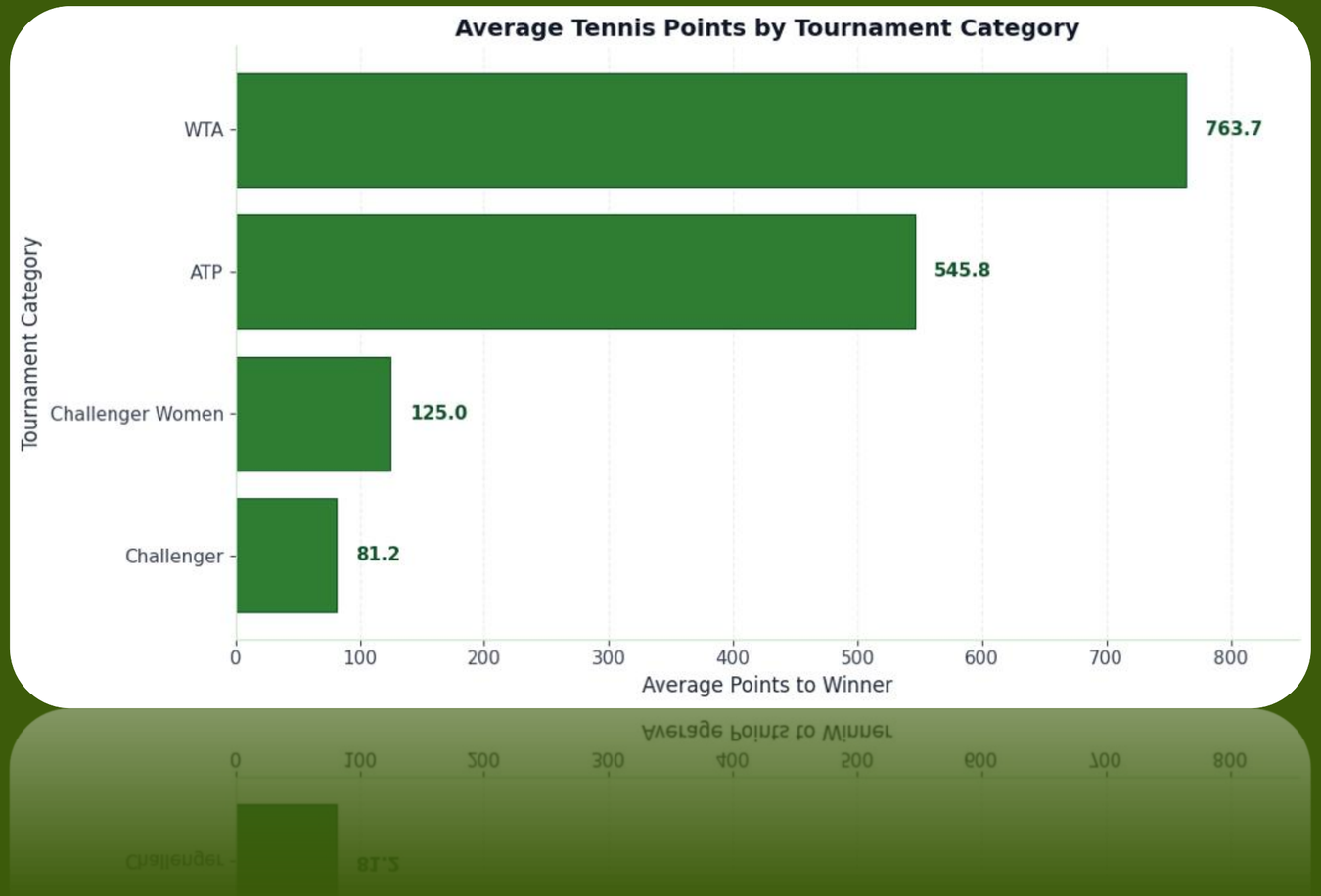
QUESTION 1: What is the average number of double faults per match?

In this tennis dataset with 11,389 matches, players make an **average of 6 double faults per match (exactly 6.05 when both players are added).** In half the matches, there are 5 or fewer double faults total, showing most games have few serving errors. Some matches have none, but the highest is 36—which is rare and probably from long, tough games. This average is normal for pro tennis: double faults happen, but players try hard to avoid them since they give free points to the opponent. Overall, about 6 per match means serves are mostly good, but pressure can cause mistakes!





QUESTION 2: Which tournament category has the highest average tennis points?

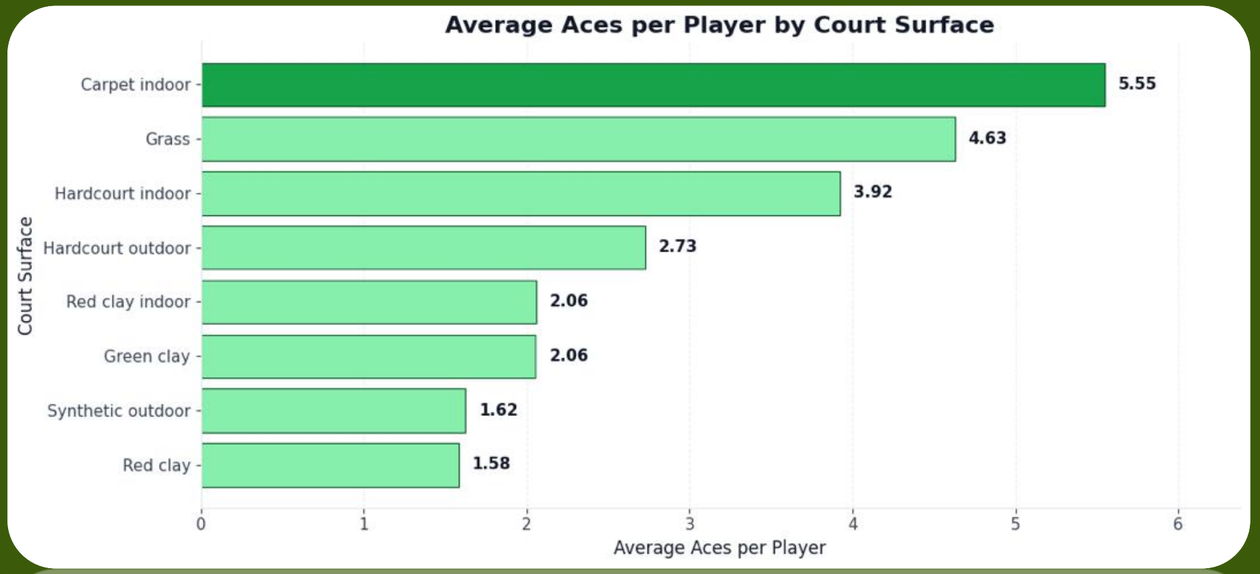


Tournament Category	Average Points
WTA	763.69
ATP	545.84
Challenger Women	125.00
Challenger	81.22

Among the tournament categories analyzed, **the WTA** category has the highest average tennis points, with an average of 763.69 points, surpassing **ATP tournaments** at 545.84 points. In contrast, Challenger and Challenger Women tournaments award significantly fewer points on average, reflecting their role as lower-tier competitions designed for player development and ranking progression. This result highlights that top-tier professional tournaments, particularly in the WTA circuit, offer the greatest opportunities for players to earn ranking points and advance their competitive standing.

QUESTION 3: On which court surface do players serve the most aces on average?

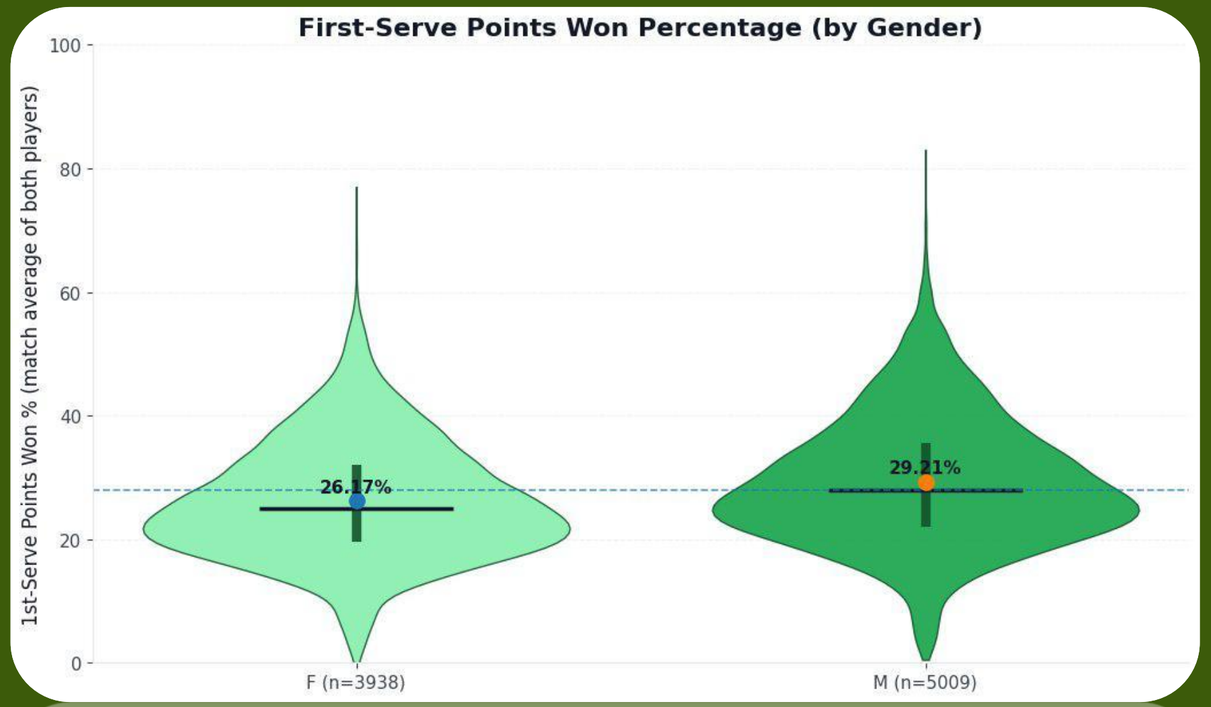




Ground Type	Average Value	Match Count
Carpet indoor	11.09	120
Grass	9.26	156
Unknown	8.52	153
Hardcourt indoor	7.85	2,068
Hardcourt outdoor	5.45	5,160
Red clay indoor	4.12	191
Green clay	4.11	18
Synthetic outdoor	3.24	78
Red clay	3.16	3,445

The data shows clear differences in the average match metric (mean) across court surfaces. **Carpet indoor courts have the highest average value (11.09)**, followed by **grass courts (9.26)**, suggesting that matches on these faster surfaces tend to be longer or involve more of the measured events. **Hardcourt indoor (7.85)** and **hardcourt outdoor (5.45)** fall in the mid-range, reflecting their balanced playing characteristics and high usage, especially outdoors with over **5,000 matches**. In contrast, **clay surfaces**, particularly **red clay (3.16)** and **green clay (4.11)**, show the lowest averages, likely due to slower play and different rally dynamics. Overall, this highlights how **surface type strongly influences match characteristics**, with faster surfaces producing higher average values and clay courts showing more conservative patterns.

QUESTION 4: What is the average first serve percentage per match, and does it differ by gender?

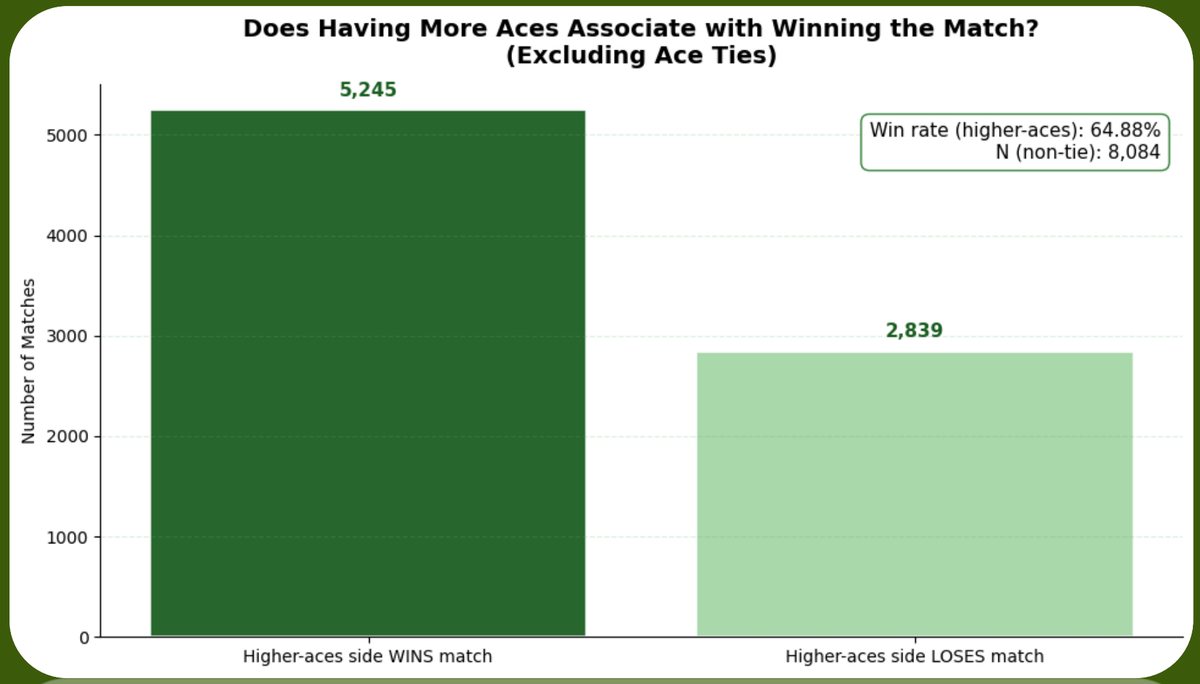


players win an average of 27.87% of points on their first serve (median 26.50%). This metric reflects how effective the first-serve phase is: once the first serve lands, how often the server converts that point into a win.
By gender, men's matches average 29.21% (median 28.00%, n=5,009), while women's matches average 26.17% (median 25.00%, n=3,938). The mean gap is 3.04 percentage points (M − F). Overall, the results suggest that first-serve point outcomes do differ by gender in this dataset, though the size of the gap should be interpreted as a descriptive difference rather than causation.



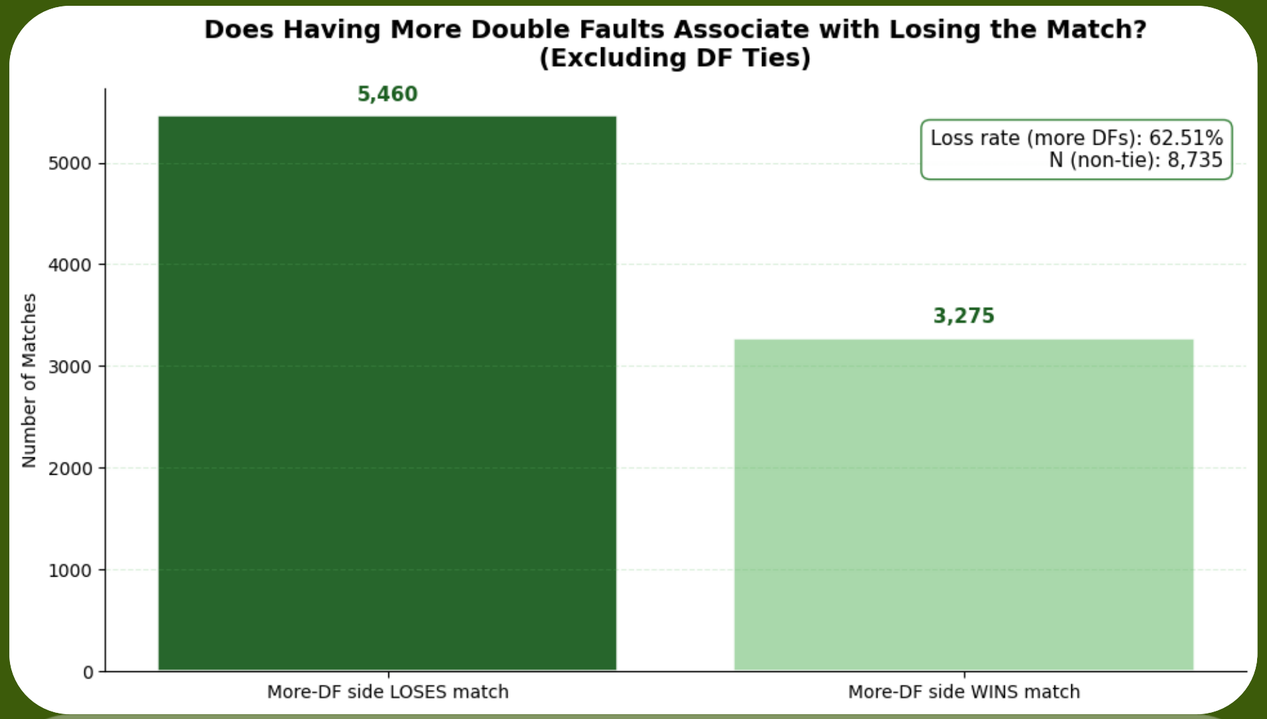


QUESTION 5: Do more aces associate with winning the MATCH?



The analysis of 10,510 professional tennis matches reveals a clear positive association between serving more aces and winning the match. **In the 8,084 matches where one player served more aces than the other (excluding 2,426 ties where both served the same number), the player with the higher ace count won 64.88% of the time.** This well-above-50% win rate indicates that having a stronger serving performance, as measured by aces, is meaningfully linked to match victory — though this reflects association, not direct causation.

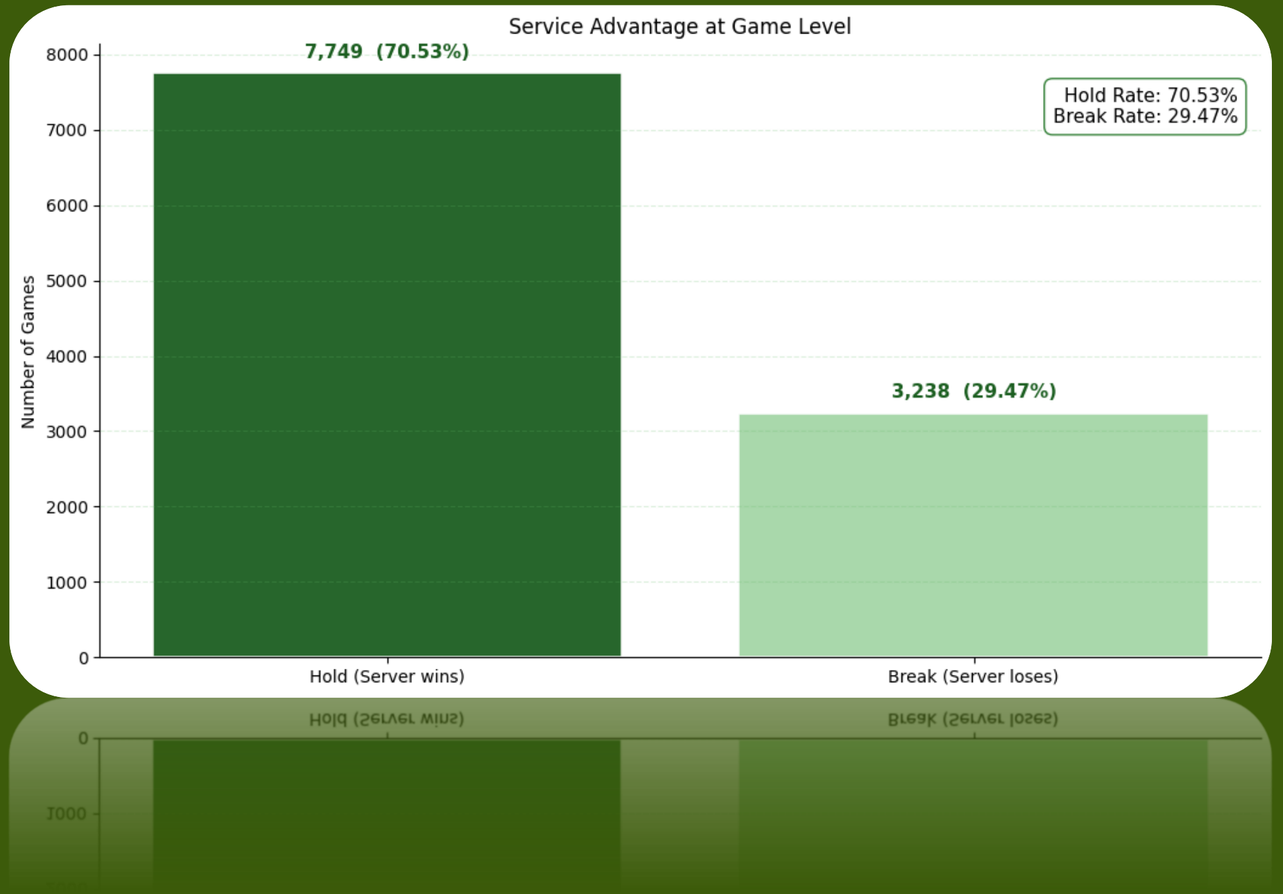
QUESTION 6: Does Committing More Double Faults Increase the Chance of Losing a Match?



The analysis of 10,510 professional tennis matches reveals a clear association between double faults and match outcomes. In the 8,735 matches where one player committed more double faults than their opponent (excluding 1,775 ties), **the player with fewer double faults won 62.51% of the time**. This notable advantage above 50% indicates that having more double faults is linked to a higher probability of losing the match. Double faults represent unforced errors that directly concede points, often disrupting serve rhythm and shifting momentum to the opponent. While this is an association rather than direct causation (other factors like return aggression or overall consistency also influence results), the data consistently shows that the player who better controls their serve—by committing fewer double faults—gains a meaningful edge in professional tennis.

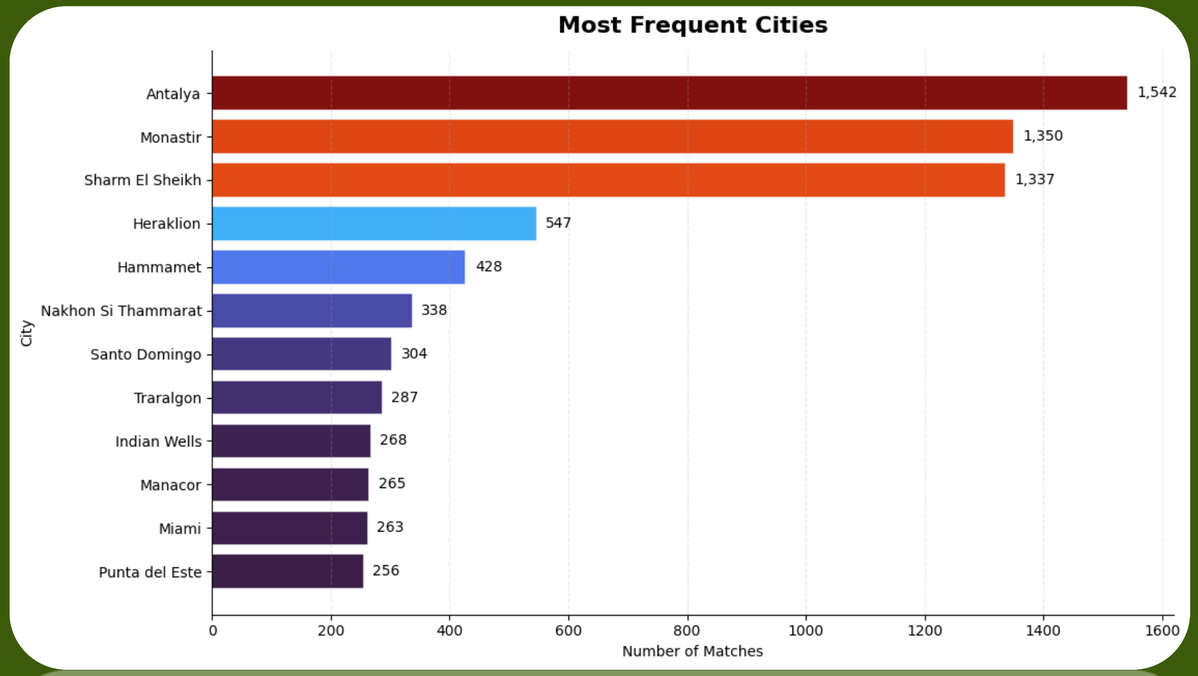
QUESTION 7: Does a Stronger Service Game Lead to More Match Wins?

The analysis of 10,987 individual service games across professional tennis matches shows a strong association between holding serve and winning the game. Servers successfully held their service game in **7,749 cases (70.53%),** while the returner broke serve in **3,238 cases (29.47%).** This means that, on average, the player serving wins their own service game more than 70% of the time — a clear and substantial advantage. Holding serve is the foundation of tennis strategy: it allows players to maintain control, reduce pressure, and force opponents to take risks on return. While breaks do occur (nearly 30% of games), the data confirms that a reliable service game is strongly linked to winning more games overall, highlighting why elite players prioritize serve consistency and power in professional play.



QUESTION 8: where matches are most frequently played?

Matches are most frequently played in **Antalya**, which stands out with 1,542 matches, making it the most active host city in the dataset. This is followed by Monastir (1,350 matches) and Sharm El Sheikh (1,337 matches), indicating that these locations serve as major hubs for hosting tennis tournaments. The high match counts in these cities suggest they regularly host multiple events, often at the Challenger or ITF level, and provide consistent playing conditions and infrastructure. Overall, the concentration of matches in a few cities highlights the importance of specific tournament centers in sustaining professional tennis competition throughout the season.

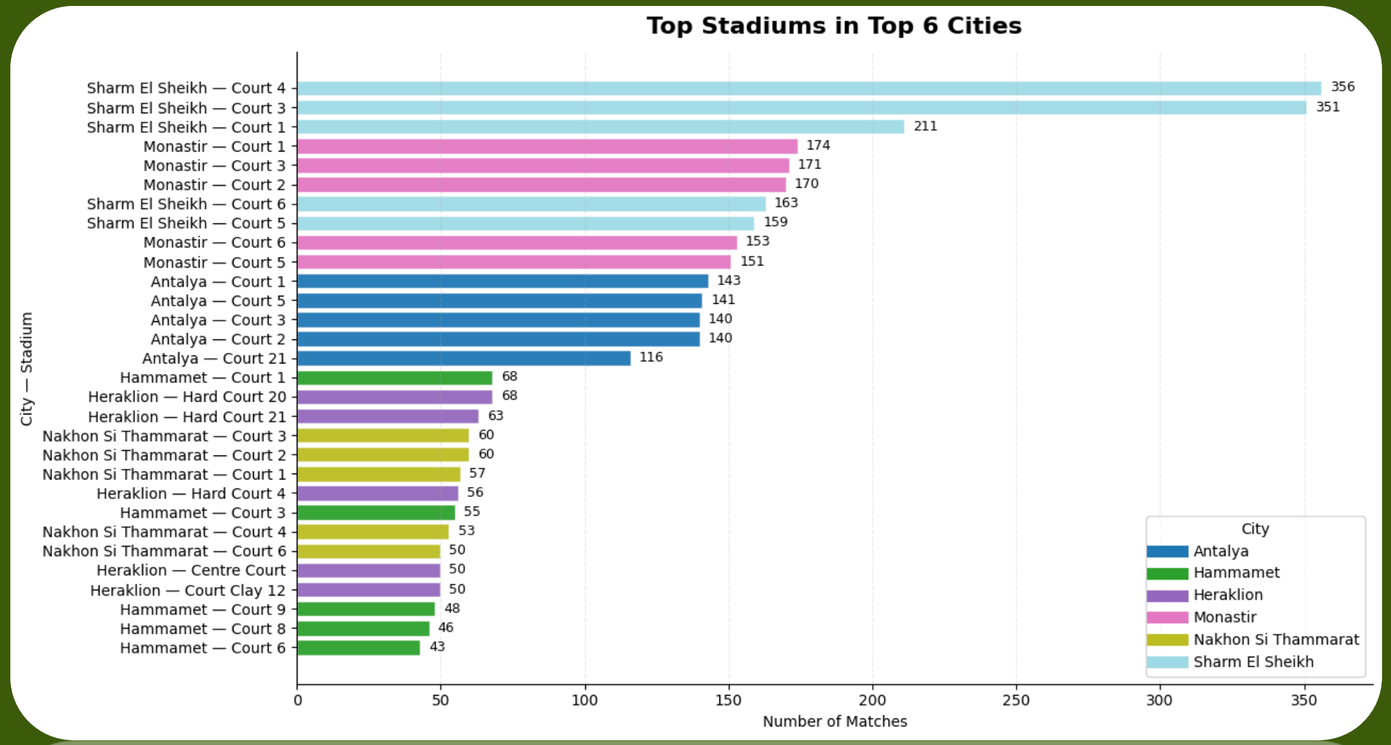


City	Number of Matches
Antalya	1,542
Monastir	1,350
Sharm El Sheikh	1,337
Heraklion	547
Hammamet	428
Nakhon Si Thammarat	338
Santo Domingo	304
Traralgon	287
Indian Wells	268
Manacor	265
Miami	263
Punta del Este	256

QUESTION 8: where matches are most frequently played?

City	Stadium / Court	Matches
Antalya	Court 1	143
Antalya	Court 5	141
Antalya	Court 2	140
Antalya	Court 3	140
Antalya	Court 21	116
Hammamet	Court 1	68
Hammamet	Court 3	55
Hammamet	Court 9	48
Hammamet	Court 8	46
Hammamet	Court 6	43
Heraklion	Hard Court 20	68
Heraklion	Hard Court 21	63
Heraklion	Court Clay 12	50
Heraklion	Centre Court	50
Monastir	Court 1	174
Monastir	Court 3	171
Monastir	Court 2	170
Monastir	Court 6	153
Monastir	Court 5	151
Nakhon Si Thammarat	Court 2	60
Nakhon Si Thammarat	Court 3	60
Nakhon Si Thammarat	Court 1	57
Nakhon Si Thammarat	Court 4	53
Nakhon Si Thammarat	Court 6	50
Sharm El Sheikh	Court 4	356
Sharm El Sheikh	Court 3	351
Sharm El Sheikh	Court 1	211
Sharm El Sheikh	Court 6	163
Sharm El Sheikh	Court 5	159

The data shows that **Sharm El Sheikh** hosts the most matches at the stadium level, with **Court 4 (356 matches)** and **Court 3 (351 matches)** being the busiest individual courts in the dataset. This indicates that Sharm El Sheikh operates as a major tennis hub with high court utilization and frequent tournament scheduling. **Monastir** also demonstrates heavy usage, with several courts each hosting over **150 matches**, reflecting a well-distributed tournament structure across multiple courts



CONCLUSION



The analysis of **2,645 professional tennis players across 100 countries** highlights the **global nature of the sport**, with talent concentrated in traditional powerhouses like the **USA, Italy, France, and Russia**, but increasingly emerging from smaller or less represented nations. Matches typically follow a **best-of-3 sets format**, with average match lengths slightly longer for male players. **Hardcourt and clay surfaces dominate tournaments**, while indoor, grass, and alternative courts are less frequent. Player performance metrics, such as **double faults**, show notable gender differences, and certain players demonstrate consistent dominance in tournament wins. Overall, the dataset provides valuable insights into **player distribution, match characteristics, and surface preferences**, offering a comprehensive view of professional tennis trends and competitive dynamics.

DANESHKAR

PRESENTATION: DATA ANALYSIS PROJECT





THANK YOU FOR YOUR ATTENTION